

STILLSTACK · REQUIREMENTS

StillStack requirements

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DOCUMENT CONTROL

Document ID	SSK-REQ-001	Version	0.4
Status	Draft	Date	2026-09-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/stillstack		
Source	docs/03-requirements.md @ 9813f20		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Add TRL 3 status from SSK-CAL-001; record decisions from SSK-DDR-001 (budget kept at \$250, tilt stand, aperture); targets unchanged
0.2	2026-09-24	Amish Chadha	First measurable requirements for TRL 2
0.1	2026-09-24	Amish Chadha	Initial scaffold

StillStack requirements

These requirements were checked by calculation at TRL 3 in SSK-CAL-001 v0.2. One is **not met** (R12 cost), six are at risk and one cannot be verified until TRL 4. The TRL 2 review decisions (SSK-DDR-001) kept the 1 m2 aperture and the 20 deg tilt on an adjustable stand. Under SSK-DDR-002 (decided by Amish on 2026-09-25, going with the recommendations) R8 was restated with a 150 °C rating for stages 1 and 2 plus a stagnation cover, and the R12 budget was raised from \$250 to \$320. The added materials and cover bring the parts to \$351.20, so R12 is still not met; a further response is proposed, awaiting Amish, in docs/REVIEW.md.

Table 1. Requirements and status at TRL 3.

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (SSK-CAL-001)
R1	Produce more distillate per unit area than a basin still	10 L per m ² of aperture per day or more on a clear day with 5.5 kWh per m ² of sun (design goal 13 L)	Stage energy balance calculation; later outdoor test with a single-basin still alongside	Met on paper: 10.6 L after warm-up deduction (11.3 L quasi-steady); the 13 L goal is not met
R2	Reuse the latent heat of condensation	Daily gained output ratio (latent heat in distillate divided by solar input) of 1.3 or more	Stage energy balance calculation	At risk: 1.34 quasi-steady, 1.26 after warm-up deduction
R3	Remove salt	Distillate conductivity 75 µS/cm or less (about 50 mg/L dissolved solids) from feed up to 40 g/L salinity	Calculation of carryover paths; later conductivity meter	Not verifiable at TRL 3: brine carryover must stay below 0.075 % of the distillate (8.6 mL per day)
R4	Keep brine and distillate apart	A physical air break of 10 mm or more between every wick edge and every distillate channel; no shared drain	Design review of the model	At risk: 10 mm dry breaks at ribs and rails and 20 mm at the low edge are modeled; the brine exit over the lidded distillate manifold is not yet detailed
R5	Run without power	No pumps, electronics or batteries; gravity feed from a user-filled container at most 1.5 m above ground	Design review	Met: fully passive; feed trough top at most 1.06 m (at 35 deg)
R6	Accept real feed water	Seawater or brackish water up to 40 g/L salinity after cloth or 20 µm prefiltering; feed-to-distillate ratio 2.5 or more so salt stays in solution	Salt balance calculation	At risk: brine at 66.7 g/L is well below saturation, but the wick needs a permeability of 3.0×10^{-11} m ² or more to carry the feed, which is not yet known
R7	Food-safe wetted parts	Every surface that touches distillate is food-contact rated (for example 316 stainless steel, food-grade coated aluminum, PP, HDPE or NSF/ANSI 51 or 61 silicone)	Material list review	At risk: coated aluminum is decided, but no coating product rated for wet, salty service at 80 °C (and 150 °C dry on plates 1 and 2) is selected
R8	Survive dry stagnation	No damage after an empty-wick day at 1,000 W/m ² and 35 °C ambient, in calm air or wind: materials in stages 1 and 2 (absorber to plate 2, including wicks, rails, ribs, coating and liner) rated to 150 °C or more; all other internal materials rated to 110 °C or more; a stagnation cover supplied and a shade rule in the instructions (restated under SSK-DDR-002)	Stagnation temperature calculation; material datasheets	At risk: stages 1 and 2 reach 144 °C in calm air (6 K margin); stage 3 reaches 110 °C in calm air (0 K margin); cover specified (BOM line 14); rail polymer and coating products not yet selected
R9				

ID	Requirement	Target	Verification (TRL 3 or later)	Status at TRL 3 (SSK-CAL-001)
	Household scale and portable	1.0 m ² aperture; panel mass 20 kg or less dry; carried by two people; footprint within 1.3 x 1.3 m	Massing model and mass estimate	Met: 16.7 kg dry (20.7 kg wet); footprint 1.28 x 1.16 m at 10 deg
R10	Easy maintenance	Wicks removable, rinsed and refitted by one person without special tools in 30 min or less; salt flushing no more often than once a week	Design review	At risk: wicks are bonded under the plates and the removal method is not defined
R11	Adjustable tilt	Tilt between 10 and 35 deg to suit latitudes of about 0 to 35 deg	Model check	Met: stand with rear struts of 0.53 to 0.94 m; model built at 10, 20 and 35 deg
R12	Low cost and buildable	Parts cost \$320 or less for the 1 m ² prototype (raised from \$250 under SSK-DDR-002); hand tools, sheet metal shears and a 3D printer only	Priced BOM	Not met: \$351.20, \$31.20 over budget

■ Assumptions

- Design day: 5.5 kWh per m² per day (19.8 MJ per m²) on the tilted aperture, 30 °C mean daytime ambient, light wind (1 m/s in SSK-CAL-001).
- Latent heat of vaporization: SSK-CAL-001 uses temperature-dependent values (about 2.33 to 2.36 MJ per kg over the stage range) in place of the flat 2.26 MJ per kg used at TRL 2.
- The WHO does not set a health-based limit for dissolved solids; 50 mg/L is chosen as a check that salt carryover is negligible, not as a potability limit.
- Household need of about 3 L per person per day for drinking and 2 L for cooking, so 10 to 11 L per day serves a household of two.